

Question Number	Answer	Mark
4(a)	The only correct answer is D 2 only <i>A is not correct because sperm cells are haploid and the shape decreases resistance</i> <i>B is not correct because sperm cells are haploid and the shape decreases resistance</i> <i>C is not correct because sperm cells are haploid and the shape decreases resistance</i>	(1)

Question Number	Answer	Additional guidance	Mark
4(b)	A description that makes reference to the following points: • {sperm cell / acrosome} releases {enzymes / acrosin / protease} (1) • (therefore) the {enzymes / acrosin} digest the zona pellucida (1) • (resulting in) the sperm cell membrane fusing with {egg cell / oocyte} membrane (allowing the nucleus to enter the egg cell) (1)	ACCEPT acrosomal reaction occurs ignore contains ignore jelly coat / follicle cells	(3)

Question Number	Answer	Additional guidance	Mark
4(c)(i)	A description that makes reference to three of the following points: - translation occurs on the {rough endoplasmic reticulum / rER / ribosomes} (1) - formation of {2° / 3° / 3D / correct named bonds} structure in rER (1) {polypeptide / protein} is packaged into vesicles by rER / {vesicles fuse with / protein transported to} Golgi (1) - modification occurs in Golgi (apparatus) (1)	ACCEPT correct description of translation / {1° structure / polypeptide} formed at ribosomes ignore protein synthesis ignore protein is folded ACCEPT correct modification in rER e.g. adding carbohydrate ignore modification unqualified ignore packaging by Golgi ignore enzyme ACCEPT description of a correct modification in Golgi apparatus e.g. adding carbohydrate / phosphate (group)	(3)

Question Number	Answer	Additional guidance	Mark
4(c)(ii)	An answer that makes reference to the following points: - the cortical granules fuse with cell membrane and release {enzymes / proteases} (1) (which result in) {hardening / thickening} of the zona pellucida (1) {acrosome enzymes / acrosin} can no longer {digest / penetrate} the zona pellucida (1)	ACCEPT enzymes released out of cell during cortical reaction ignore exocytosis unqualified ignore {hardening / formation} of {fertilisation membrane / jelly coat} ACCEPT {(cortical) enzymes / proteases} modify sperm {receptors/ binding sites} (so sperm can no longer bind to them) ACCEPT {(cortical) enzymes / proteases} clip the attachment proteins in the perivitelline, releasing any additional sperm that have attached	(3)